What is KVNO in Active Directory status ?

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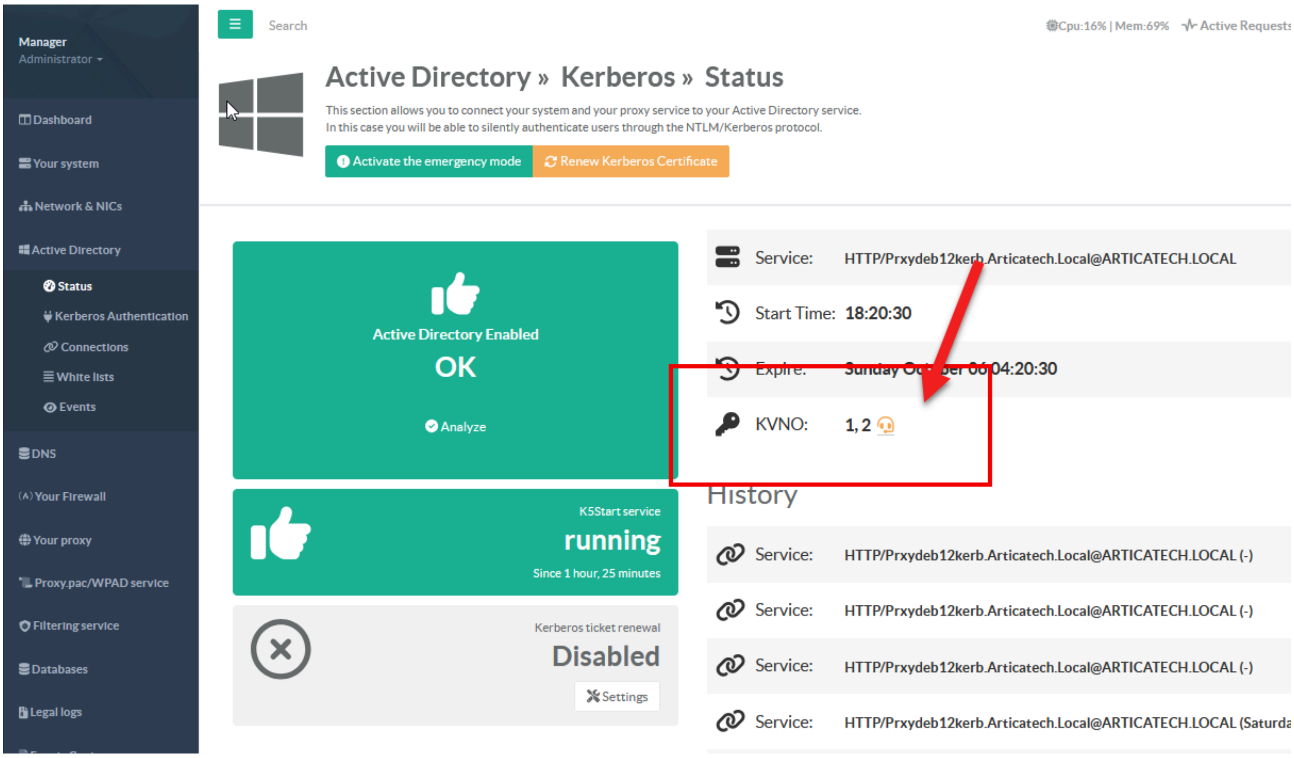
> In **Kerberos**, **KVNO** stands for **Key Version Number**.
> It is a number associated with a particular **encryption key** for a Kerberos principal (such as a user, service, or host). The KVNO helps Kerberos distinguish between different versions of a principal's key when it changes.

In the context of **Active Directory** (AD), which is Microsoft's implementation of **Kerberos** as part of its authentication system, **KVNO** (Key Version Number) plays a similarly critical role as in any Kerberos environment.

It tracks the version of encryption keys for a principal (such as users or services) and ensures that the correct key version is used when tickets are issued or validated.

> The **KVNO** is incremented by Active Directory whenever the **password for an account is changed** (including user, service, or computer accounts).



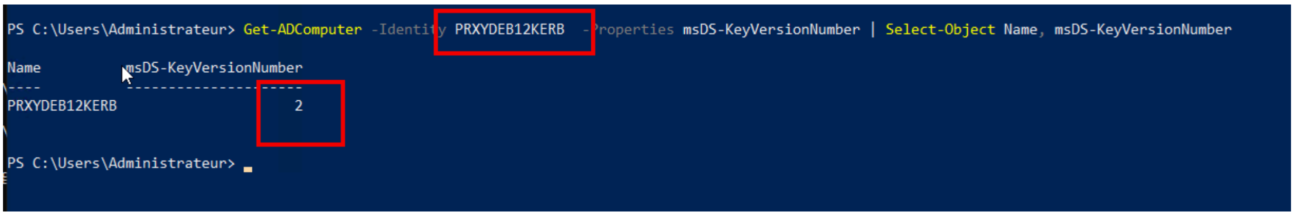
You can see the KVNO number in the Active Directory status page

- **Key Version in AD**:
  In Active Directory, each user, service, or computer account has an associated encryption key, and the **Key Version Number (KVNO)** is stored along with this key. The KVNO is used to keep track of key changes, such as when a password is changed.
- **Password or Key Changes**:
  When a **user** or **service** account password is changed in Active Directory, a new encryption key is generated for the account, and the **KVNO** is incremented. This ensures that the KDC (Key Distribution Center) and other services know which key version to use for authenticating and decrypting Kerberos tickets.
- **Kerberos Tickets**:
  When Active Directory issues a Kerberos ticket, it includes the KVNO in the ticket. The **KVNO** tells the service (or KDC) which version of the key should be used to decrypt the ticket.
- **Active Directory Service Accounts**:Service accounts (such as those used by Artica services) also have their keys stored in Active Directory. If a service's account password is changed, the KVNO for the service's account is incremented. When the service receives a Kerberos ticket, it checks the KVNO to ensure it uses the correct key version to decrypt the ticket.
- **Managing Service Principal Names (SPNs)**:
  In Active Directory, a **Service Principal Name (SPN)** is a unique identifier for services running on servers. SPNs are tied to accounts (e.g., a service account) and their associated keys. The KVNO for a given SPN helps ensure that the proper key version is used to authenticate a service.
- **Misaligned KVNO Issues**:
  When the KVNOs between the KDC and the client or service are out of sync (for instance, when a password has been changed but the service is still using an old key), Kerberos authentication can fail with errors like **KRB_AP_ERR_MODIFIED**. This happens when a service attempts to decrypt a ticket using an old key, but the ticket was encrypted with a new key (due to a KVNO mismatch).

Methods to Check KVNO in Active Directory

Use this command using PowerShell

```
Get-ADComputer -Identity [computer_account] -Properties msDS-KeyVersionNumber | Select-Object Name, msDS-KeyVersionNumber
```



This will show the key version number for that user in Active Directory